DATA ITEM DESCRIPTION

Title: BUILT-IN TEST (BIT) DATA / TEST VERTICALITY REPORT

Number: DI-SESS-82498
AMSC Number: N10594
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

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Limitation: N/A
GIDEP Applicable: N/A
Project Number: SESS-2025-025

Use/relationship: The Built-In Test (BIT) Data / Test Verticality Report documents the technical BIT design implementation in sufficient detail to perform aircraft integration and determine whether all the BIT interface requirements have been met. The intent is to assure proper BIT philosophy, hardware/software planning and design coordination, configuration management, and development of logistics capabilities (e.g., Automated Maintenance Environment (AME)/ Configuration and Fault Reporting System (CFRS), tech pubs, etc.). It also documents false alarm mitigation strategies, to guide the software design. Finally, the report includes the fault detection and fault isolation model for the equipment used in the design, assessment, and procurement specifications.

This DID contain the format, content, and intended use information for the data product resulting from the work task described in the contract Statement of Work (SOW).

Requirements:

1. Reference documents.

1.1. DI-SESS-82078A, Integrated Health Management System (IHMS) Description and Data Architecture (DDA). (available at https://quicksearch.dla.mil.)

1.2. DI-SESS-81497B, Reliability and Maintainability Predictions Report.
(available at https://quicksearch.dla.mil.)

1.3. DI-SESS-81603, Maintainability and Built in Test Demonstration Report.
(available at https://quicksearch.dla.mil.)

1.4. DI-SESS-81604B, Maintainability and Built in Test Demonstration Procedure.
(available at https://quicksearch.dla.mil.)

2. Format. This report is to be provided in an electronic format. The BIT Fault Detection (FD)/Fault Isolation (FI) worksheet is to conform to the provided format, and any modifications to the content or format defined therein require written NAVAIR approval. Additional columns or data may be added to facilitate computations as desired.

3. Content.

3.1. EXTERNAL DOCUMENT REFERENCES. The following ground rules are to be observed for referencing external documents. References for external documents are only permitted for the Built-in Test Design Requirements Data section of this report. The inclusion of external documents via appendices to this report is preferred. If the applicable data of an external reference can be condensed to less than five (5) pages, its text

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should be contained in this report. Where applicable to items in this report, specific paragraphs, tables, figures, etc., should be referenced instead of global references to external documents. Only direct references are to be used; reference documents should not "chain-reference" other reference documents. A preface summary table of references, listing all external references used, including revision letters and dates, and specific paragraphs, figures, tables, appendices, etc., is also to be provided for this report. Prior to being referenced by this report, each externally referenced document must have been delivered to NAVAIR, preferably in electronic format.

3.2. Description.

3.2.1. BUILT-IN-TEST (BIT) DESIGN REQUIREMENTS DATA. This section of the report documents the technical BIT design implementation in sufficient detail to perform aircraft integration and determine whether all the BIT interface requirements have been met. The intent is to assure proper BIT philosophy, hardware/software planning and design coordination, configuration management, and development of logistics capabilities (e.g., AME/CFRS, tech pubs, etc.). At a minimum, this section includes the following information:

3.2.1.1. A general description of the avionics hardware, BIT specification requirements, and avionics BIT design philosophy. This description addresses system operation, system-level BIT hardware/software allocations, and BIT hardware test signals. Include descriptions of any BIT-induced system inhibit or interrupt commands along with justification for implementation if required. If applicable, provide a detailed description of all system calibrations and system-level BIT tests.
DATA DELIVERY: INITIAL, update at INTERMEDIATE and update at FINAL.

3.2.1.2. A description of the system operation and system maintenance support concept at the organizational, intermediate, and depot levels of maintenance is to be provided. Include, at a minimum, the role of BIT and support equipment at each level.
DATA DELIVERY: INITIAL, update at INTERMEDIATE and update at FINAL.

3.2.1.3. A detailed description of the power subsystem, including power recycling, overload protection features, and all self-test features, as applicable.
DATA DELIVERY: INTERMEDIATE and update at FINAL.

3.2.1.4. A family tree drawing for each Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) graphically illustrating the hierarchical relationship between its assemblies is to be provided. The drawing(s) clearly shows the composition of the WRA / LRU from highest to lowest level of assembly down to and including the sub-Shop Replaceable Assembly (sub-SRA) / Shop Replaceable Unit (SRU) level, with the name and part number of each assembly identified.
DATA DELIVERY: INITIAL, update at INTERMEDIATE, and update at FINAL.

3.2.1.5. Data describes the design of each Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) and Shop Replaceable Assembly (SRA) / Shop Replaceable Unit (SRU), including a block diagram, a description of assembly operation, and a detailed description of Built-In Test (BIT) features. For SRAs / SRUs with embedded self-test (e.g., Autonomous Self-Test, Built-In Self-Test, etc.), include a description of the BIT coverage achieved as well as the method for executing the self-test under the applicable system BIT modes.

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3.2.1.6. Provide schematic data that is easily traceable to the information contained in the Built-In Test (BIT) analysis worksheets defined in Part C of this document. These schematics are to be marked to identify the functions

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within the Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) and to identify the BIT tests that test each function.
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3.2.1.7. A description of all aircraft BIT and maintenance interfaces, including the following:
DATA DELIVERY: INITIAL, update at INTERMEDIATE and update at FINAL.

3.2.1.7.1. Justification for usage of any physical BIT indications (e.g., LEDs, mechanical BIT indicators).

3.2.1.7.2. A block diagram and technical description of the avionics system's BIT hardware/software interface to the aircraft hardware/software.

3.2.1.7.3. Technical descriptions and illustrations of all BIT displays for tactical mode assessment and degraded performance (if applicable), maintainer BIT data, and fault isolation results.

3.2.1.7.4. The organization of all BIT matrix data provided to the Mission Computer over the MIL-STD-1553 multiplex bus (or equivalent). A BIT failure code cross-reference table identifying the BIT modes and tests generates the following:

3.2.1.7.4.1. 1553 System/Sub-system Fail Codes (e.g., Go/No-Go, Over-temperature, Checksum status).

3.2.1.7.4.2. Operational Flight Program (OFP) Error Codes (e.g., OFP recognized errors/failures to be included in the fault log).

3.2.1.7.4.3. Weapon Replaceable Assembly (WRA)/ Line Replaceable Unit (LRU) – Identifies the System WRA (LRU) failure status BIT set (if required by Configuration Identification and Data System (CIDS)).

3.2.1.7.4.4. BIT Modes – Identifies the BIT mode of the test that generates the failure code Periodic Buit-In Test (PBIT) or Initiated Built-In Test (IBIT).

3.2.1.7.4.5. Section (B) BIT Description Paragraph Number – Indicates the location of the test summary, addressing the fail code and the corresponding BIT test.

3.2.1.7.5. A detailed description of all non-volatile fault log data, including:

3.2.1.7.5.1. Memory locations of stored data.

3.2.1.7.5.2. Hardware/software interfaces and methods of data retrieval.

3.2.1.7.5.3. Parameters stored (pass/fail criteria, tolerances) and correspondence to specific test numbers.

3.2.1.7.5.4. Other available information (e.g. calibration data) relevant to O-level/depot diagnostics.

3.2.1.7.5.5. When applicable, a complete listing (with pass/fail criteria) of all BIT test and calibration measurement data available for supplemental diagnostic- Diagnostic Test Module (DTM) use along with procedure

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for data retrieval.

3.2.1.8. Descriptions of each applicable BIT mode (e.g., Power-up BIT (PUBIT) or Start-up BIT (SBIT)), Background/Periodic BIT (BBIT or PBIT), Initiated BIT (IBIT), Maintenance BIT (MBIT)), including a flowchart of BIT test sequences and timelines. Include detailed test descriptions to define the test objective, test mechanization, and method for setting the test threshold for each Built-In Test (BIT). Identify the relationship of test tolerances to hardware requirements being tested, along with pass/fail criteria for each BIT. Additionally, identify the relationship between the BIT test levels/tolerances and the corresponding Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) and Shop Replaceable Assembly (SRA) / Shop Replaceable Unit (SRU) depot test levels/tolerances.
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3.2.1.9. Detailed descriptions of each test executed during at least one BIT mode, including definition of the test objective, test mechanization and method for setting the test. Identify the relationship of test tolerances to hardware requirements being tested, along with pass/fail criteria.
DATA DELIVERY: INTERMEDIATE and update at FINAL.

3.2.1.10. For non-digital tests relationships between the test levels/tolerances of BIT and the corresponding off-aircraft black box/module tests. .
DATA DELIVERY: INTERMEDIATE and update at FINAL.

3.2.1.11. A detailed description of all false alarm filtering (i.e., “M” of “N” event or persistence counting) and mitigation techniques used.
DATA DELIVERY: INITIAL, update at INTERMEDIATE and update at FINAL.

3.2.1.12. Recommended mechanisms and plan for adjusting BIT accept/reject and false alarm filtering thresholds, for pre-production and production equipment.
DATA DELIVERY: INTERMEDIATE, and update at FINAL.

3.2.1.13. Provide a description to address the recording and evaluation of BIT test results by BIT fault detection, fault isolation and mode assessment software. Flowcharts and/or tables depicting the logic used by BIT to achieve unambiguous fault isolation for the system.
DATA DELIVERY: INTERMEDIATE and update at FINAL.

3.2.1.14. Technical justification for any noncompliance of BIT performance, along with achievable levels.
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3.2.1.15. For any failures or group of failures identified in Section C, as undetectable or poorly isolated, other than excluded or no effect, provide rationale justifying lack of coverage and recommend alternative detection (e.g., potential aircraft level BIT tests, pilot observations, etc.). Design approaches or trade-offs that were considered to improve fault coverage should be described, along with the rationale for their rejection. Identify advantages (e.g., reduced cost, reduced false alarm susceptibility, etc.) that may be gained from tests using other interfacing avionics.

DATA DELIVERY: INTERMEDIATE and update at FINAL.

3.2.1.16. A description of diagnostic advantages that may be gained from tests using other interfacing avionics.
DATA DELIVERY: INTERMEDIATE and update at FINAL.

3.2.1.17. Justification for uses of a WRA (LRU) failure indicator other than that specified.
DATA DELIVERY: INTERMEDIATE and update at FINAL.

3.2.1.18. The time required determining a go or no-go condition of the equipment in each of the BIT operating modes.
DATA DELIVERY: INTERMEDIATE and update at FINAL.

3.3. BIT FALSE ALARM DESIGN MITIGATION (FADM) ANALYSIS. This section of the document provides a qualitative false alarm mitigation analysis for the equipment. The analysis is to be proactively used to guide the hardware and software design, mitigating susceptibility to Built-In Test (BIT) False Alarms and Can Not Duplicates (CNDs). The following “lessons-learned” checklist is to be used to assess each black box and module design, as well as the overall design, documenting the hardware and software features provided by each. The Supplier is encouraged to incorporate additional field or factory lessons learned that may apply to the design.
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FALSE ALARM MITIGATION CHECKLIST
Software Design BIT Concerns

POTENTIAL FALSE ALARM CHARACTERISTICS/CAUSES	SUSCEPTIBILITY (H, M, L or N/A)	COMMENTS/MITIGATION EFFORTS
Are power-cycling issues such as interrupts, rise times after turn on, decay times after turn off and stabilization periods sufficiently filtered or accounted for in BIT?		
Are signal timing/loading characteristics stabilized with respect to minimum, nominal and maximum supply voltages before being assessed by BIT?		
Does the termination of IBIT prior to its completion produce signal transients?		
Does BIT wait until sufficient equipment warm-up has occurred?		
Does appropriate logic initialization occur before BIT is started?		
Are random and sinusoidal vibrations properly filtered to avoid BIT false alarm?		.
Are thermal limits as specified in the specification accounted for in the BIT measurement window?		.
Is vertical testability fully integrated in the BIT design using appropriate tolerance budgeting levels?		
Are BIT fault messages clear and concise to prevent potential maintenance induced errors?		
Are BIT threshold and filtering parameters easily changed with software?		
Does the design preclude BIT software from trying to operate hardware outside hardware margins?		
Does design mitigate BIT software from reporting performance anomalies as failures when the hardware is operating under environmental conditions beyond those specified for its performance?		

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FALSE ALARM MITIGATION CHECKLIST - Continued
Software Design BIT Concerns

POTENTIAL FALSE ALARM CHARACTERISTICS/CAUSES	SUSCEPTIBILITY (H, M, L or N/A)	COMMENTS/MITIGATION EFFORTS
Does design provide for the following? - non-volatile collection and storage of avionics diagnostic and calibration data - aircraft time, operation and environment data - critical box temperature, elapsed time and serial number - other data vital to the effective support of the maintenance concept, including warranty administration		
Are BIT thresholds consistent with hardware specifications and adequately documented?		
Have BIT software changes been verified with all hardware configurations?		
Is the software timeline < 100% and adequate to support reliable system operation and future growth?		
Does mode switching or mode control affect signals measured by BIT?		
Will the absence of equipment interface signals cause a BIT failure? (e.g., Internal Navigation System (INS), Weight of Wheels (WOW))		
Is EMI susceptibility accounted for in BIT software?		
Has a proven or accepted method for fault filtering been implemented to prevent false alarms?		
Is software tolerance and margin analysis complete?		

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FALSE ALARM MITIGATION CHECKLIST - Continued
Hardware Design BIT Concerns

POTENTIAL FALSE ALARM CHARACTERISTICS/CAUSES	SUSCEPTIBILITY (H, M, L or N/A)	COMMENTS/MITIGATION EFFORTS
Was circuit trace layout considered with respect to placement and trace length to avoid crosstalk and ensure noise immunity? Are signal routing conditions present on Circuit Card Assemblies (CCAs) or wiring, which may lead to false triggering or signal degradation?		
Is BIT for Very-Large-Scale Integration (VLSI)/ Very High-Speed Integrated Circuit (VHSIC)/ Application-Specific Integrated Circuit (ASIC) devices sufficient to detect the propagation of internal faults?		
Do the component stress and thermal analyses indicate that component rating and selection, module component placement and thermal design, and aircraft cooling provisions are adequate to ensure consistent hardware operation and stable signal parameters under all operational scenarios?		
Does the design possess adequate power and ground plane noise immunity? Is the design sufficiently shielded, filtered and partitioned to preclude false alarms resulting from internally or externally-generated electromagnetic or radio frequency interference?		
Is alignment necessary for any circuitry after module replacement?		
Are tolerance and margin (extreme value, Monte Carlo, etc.) analyses complete for all hardware BIT test thresholds? Do test tolerance and margin analyses adequately support a “cone of tolerance” approach (progressively tighter test tolerances from system to black box to module)?		
If required, is module fault isolation by BIT adequate to enable compliance with the specified off-aircraft fault isolation requirement?		

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FALSE ALARM MITIGATION CHECKLIST - Continued
Hardware Design BIT Concerns

POTENTIAL FALSE ALARM CHARACTERISTICS/CAUSES	SUSCEPTIBILITY (H, M, L or N/A)	COMMENTS/MITIGATION EFFORTS
Does BIT test strategy and fault coverage ensure test verticality and test traceability between System, black box and module testing?		
Are connector, Printed Wiring Board (PWB) and rigid wire harness technology/quality and wire/cable routing sufficient to assure protection from intermittent connections during flight?		
Are assembly mounting, vibration and strength provisions adequate to mitigate intermittent connections during flight?		

3.3. BIT FD/FI AND TEST VERTICALITY ANALYSIS**.** This section of the document provides a quantitative, failure rate-weighted BIT fault detection/fault isolation (FD/FI) and test verticality model for the equipment, to be used in both design and assessment of BIT performance capabilities. This analysis addresses black box and module level Fault Detection/Fault Isolation (FD/FI) as described in the Procurement specification. The Built-In Test (BIT) FD/FI and test verticality assessment, along with supporting data, are to include the following elements and conform to the specified worksheet format. Any modifications to the content or format defined herein require written NAVAIR approval. Additional columns may be added to facilitate computations as desired. The general approach is to assess the BIT and off-aircraft coverage for each circuit card/module when installed in the aircraft system, then combine those results into an assessment of each black box or system, as applicable.

3.3.1. GROUND RULES AND ASSUMPTIONS

3.3.1.1. The following ground rules and assumptions are to be used in preparing the BIT analysis worksheets. Ground rules and assumptions can be added or deleted subject to written NAVAIR approval.

3.3.1.1.1. Component failure rates are to be taken from the Reliability Prediction.

3.3.1.1.2. Each component failure mode and its effect on normal system operational performance are evaluated independently of any other component failure modes (a single-point failure philosophy).

3.3.1.1.3. The analysis includes only failure modes of electronic/electrical component (circuitry), Radiofrequency (RF) cabling, and electromechanical components. Therefore, connectors, PCBs, solder connections and wiring are excluded between Shop Replaceable Assemblies (SRAs) / Shop Replaceable Units (SRUs) and external Weapon Replaceable Assemblies (WRAs) / Line Replaceable Units (LRUs) are excluded

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3.3.1.1.4. Failure modes of components used for the following purposes, that do not affect normal system operational performance, are excluded. Normal system operational performance means that all functional circuits, safety circuits, and BIT circuits are fully capable of performing their intended functions.

3.3.1.1.4.1. Failure modes of components employing "good engineering practices" (e.g. redundant pull-up resistors, decoupling or filtering capacitors, etc.) which do not affect normal system operational performance are excluded.

3.3.1.1.4.2. Failure modes of components dedicated to test interface circuitry (e.g., software debug, signature/ configuration. resistors, test point isolation resistors, other circuits for depot test use only, etc.) which do not affect normal system operational performance are excluded.

3.3.1.1.4.3. Failure modes of components provided specifically for safety circuitry required to protect the depot maintainer, that do not impact normal system operational performance are to be classified as “no effect” for two-level OEM Depot programs.

3.3.1.1.4.4. All spare/growth circuitry that does not impact normal system operational performance, is to be classified as “no effect”.

3.3.1.1.5. Failure modes of circuitry that cause a power shutdown or a failure to communicate with the external system will render the WRA (LRU) inoperable ("Dead Box"). Credit may be taken for BIT fault detection and isolation, if the external system (i.e. Mission Computer) does the following:

3.3.1.1.5.1. Detects the WRA (LRU) or system failure (dependent on having a "Ready" signal provided to Mission Computers (MC)).

3.3.1.1.5.2. Reports the corresponding loss of functionality to the pilot.

3.3.1.1.5.3. Reports the faulty black box to the ground crew (substantiation of how the fault is isolated is required).

3.3.1.1.6. A test is defined as a specific BIT or off-aircraft measurement/observation that verifies one or more attributes of the system.

3.3.1.1.7. Failures of Built-In Test (BIT) circuitry that only affect BIT operation are to be included in the analyzed failure rate unless specifically excluded by the Performance/Procurement Specification.

3.3.1.1.8. Overload and protection circuitry that does not contribute to the normal operational system is excluded (unless failure of circuitry would cause a functional failure).

3.4. DEFINITIONS

3.4.1. FAULT DETECTION

Calculate fault detection for detection by Power-up/Start-up Built-In Test (PUBIT), Background/Periodic Built-In Test (PBIT), and the combined BIT modes (PBIT + Initiated Built-In Test (IBIT)) for both the

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Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) and Shop Replaceable Assembly (SRA) / Shop Replaceable Unit (SRU) if required. The total analyzed failure rate detected by PUBIT, PBIT and the combined BIT modes (PBIT + IBIT) is listed on the WRA (LRU) worksheet

Fault detection is defined as:

$$\% \text{ Fault Detection} = \frac{\text{Total Detected Failure Rate}}{\text{(100) Total Analyzed Failure Rate}}$$

Notes

Total Analyzed Failure Rate = Total Failure Rate – Excluded Failure Rate. The Total Analyzed Failure Rate can be obtained from the spreadsheet directly

Total Detected Failure Rate can be obtained from the bottom of each column under detected failure rate for PUBIT, PBIT and the combined BIT modes.

NOTE: Total Detected Failure Rate = Total Analyzed Failure Rate – Undetected Failure Rate

3.4.2. ISOLATION

Calculate fault isolation for unambiguous isolation to one (1) Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU). The WRA (LRU) summary worksheets summarize this data.

Fault Isolation is defined as:

$$\% \text{ Fault Isolation} = \frac{\text{Total Isolated Failure Rate}}{\text{(100) Total Detected Failure Rate}}$$

Notes

Total Detected Failure Rate via all BIT modes (PBIT + IBIT) can be obtained from the WRA (LRU) -level spreadsheet directly.

Total Detected (PBIT + IBIT) Failure Rate isolated to one (1) WRA (LRU) by BIT can be obtained from the WRA (LRU) –level spreadsheet directly.

3.4.3. O-LEVEL FAULT DETECTION

Calculate fault detection for Organizational-level (O-level) detection by any method for the Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU). The total analyzed failure rate (Verticality) and Overall BIT/O-Level Detection data is listed on the WRA (LRU) worksheet. Fault detection is defined as:

$$\% \text{ Fault Detection} = \frac{\text{Total Overall BIT/O-Level Detected Failure Rate}}{\text{(100) Total Analyzed Failure Rate (Verticality)}}$$

Notes

Total Analyzed Failure (verticality) can be obtained from the spreadsheet directly

Total Detected Failure Rate (verticality) can be obtained from the spreadsheet directly

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3.4.4. DEPOT WRA (LRU) VERTICALITY

Calculate Depot Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) verticality for the WRA / LRU. The total analyzed failure rate (Verticality) and Depot detection data is listed on the WRA (LRU) worksheet.

Verticality is defined as:

$$\% \text{ Depot Verticality} = \frac{\text{Total Depot Verticality Failure Rate}}{\text{(100) Total Analyzed Failure Rate (Verticality)}}$$

Notes
Total Analyzed Failure Rate (verticality) can be obtained from the spreadsheet directly
Total Depot Verticality Failure Rate can be obtained from the spreadsheet directly

3.4.5. ATP WRA (LRU) VERTICALITY

Depot ATP verticality must be calculated for the WRA (LRU). The total analyzed failure rate (Verticality) and ATP detection data is listed on the WRA (LRU) worksheet.

Verticality is defined as:

$$\% \text{ Depot Verticality} = \frac{\text{Total ATP Verticality Failure Rate}}{\text{(100) Total Analyzed Failure Rate (Verticality)}}$$

Notes
Total Analyzed Failure Rate (verticality) can be obtained from the spreadsheet directly
Total ATP Verticality Failure Rate can be obtained from the spreadsheet directly

3.5. CIRCUIT FAILURE CLASSIFICATION

All components are to be classified per Table 1. Classes 4 and 5 are excluded from the analysis calculations (but not from the worksheets) because they do not impact the mission

TABLE 1

CLASS	DESCRIPTION OF CIRCUIT	ANALYSIS IMPACT
1	Functional Circuit	Operational/Mission
2	Equipment/Operator Safety	Operational/Mission
3	Dedicated BIT Hardware	Operational/Mission

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4	Dedicated Test Interface	GSE Use Only/No Effect
	a) Signature/Configuration Resistor	
	b) Test Point Isolation Resistor	
	c) GSE Interface - Components for test use whose failure cannot cause degradation of the system	
5	Non-Mission/No Effect Components	Good Design Practice/
	a) Open pull-up resistors	No Effect
	b) Open filter capacitors	
	c) Shorted Inductors (except when performance is affected)	
	d) Hysteresis Resistors	
	e) Unused components or component elements	
	f) Additional - case by case	
6	Other Contractually Excluded Hardware	No Effect

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3.6. FAILURE MODE ALLOCATION/DISTRIBUTION OF COMPONENTS

The allocation/distribution of component failure modes to be used in the BIT analysis is shown in Table 2. Any deviation and/or addition from this table requires NAVAIR concurrence.

TABLE 2

DEVICE TYPE	FAILURE MODE	PERCENTAGE	REFERENCE GUIDE
CAPACITOR, FIXED CERAMIC DIELECTRIC	SHORT CHANGE OF VALUE OPEN	49 29 22	FMD-91
CAPACITOR, MICA/GLASS	SHORT CHANGE OF VALUE OPEN	72 15 13	FMD-91
CAPACITOR, TANTALUM ELECTROLYTIC	SHORT CHANGE OF VALUE OPEN	69 14 17	FMD-91
CRYSTAL UNIT	OPEN NO OSCILLATION	89 11	FMP-91
COIL	SHORT OPEN CHANGE OF VALUE	42 42 16	FMD-91
DIODE, GENERAL	SHORT OPEN CHANGE OF VALUE	49 36 15	FMD-91
DIODE, ZENER, VOLTAGE REGULATOR	SHORT OPEN CHANGE OF VALUE	20 45 35	FMD-91
DIODE, LIGHT EMITTING	SHORT OPEN	30 70	FMD-91
DIODE, RECTIFIER, SILICON CONTROLLED	SHORT OPEN	98 2	FMD-91
DIODE THYRISTOR	SHORT OPEN STUCK ON STUCK OFF	40 10 5 45	FMD-91
DIODE, TRIAC	STUCK ON STUCK OFF	10 90	FMD-91
MICROCIRCUIT, DIGITAL, BIPOLAR	OUTPUT STUCK HIG OUTPUT STUCK LO OUTPUT (OPEN) INPUT OPEN	28 28 22 22	FMD-91

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TABLE 2 - Continued

MICROCIRCUIT, DIGITAL, MOS	OUTPUT STUCK HIGH OUTPUT STUCK LO OUTPUT (OPEN INPUT OPEN SUPPLY OPEN	08 08 36 36 12	FMD-91

DEVICE TYPE	FAILURE MODE	PERCENTAGE	REFERENCE GUIDE
MICROCIRCUIT, INTERFACE	SUPPLY OPEN OUTPUT STUCK LO OUTPUT (OPEN) INPUT OPEN	10 58 16 16	FMD-91
MICROCIRCUIT, LINEAR	IMPROPER OUTPUT NO OUTPUT	77 23	FMD-91
MICROCIRCUIT, MEMORY, BIPOLAR	SLOW TRANSFER OF DATA DATA BIT LOSS	79 21	FMD-91
MICROCIRCUIT, MEMORY, MOS	DATA BIT LOSS SHORT OPEN SLOW TRANSFER OF DATA	34 26 23 17	FMD-91
OPTO-ELECTRONIC SENSOR	OUTPUT SHORTED OUTPUT OPEN	50 50	FMD-91
MOTOR	WINDING FAILURE (OPEN) WINDING FAILURE (SHORT) BEARING FAILURE (OVERCURRENT) OPEN SLIP RING, BUSH, COMMUTATOR	20 20 45 15	MIL-HDBK-338
RELAY	CONTACT FAILURE (SHORT) CONTACT FAILURE (OPEN) OPEN COIL	30/N 45/N 25	MIL-HDBK-338
RESISTOR, FIXED CARBON AND METAL FILM	OPEN CIRCUIT CHANGE OF VALUE	80 20	MIL-HDBK-338
RESISTOR, FIXED COMPOSITION	OPEN CIRCUIT CHANGE OF VALUE	5 95	MIL-HDBK-338
RESISTOR, THERMISTOR	OPEN PARAMETER CHANGE SHORT	63 22 15	FMD-91
RESISTOR, VARIABLE	OPEN ERRATIC OUTPUT (WIPER FAILURE) SHORT	53 40 7	FMD-91
SWITCH, TOGGLE	CONTACT FAILURE (OPEN) SWITCH FAILURE (STUCK) CONTACT FAILURE (SHORT)	65/N 19 16/N	FMD-91

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TABLE 2 - Continued

DEVICE TYPE	FAILURE MODE	PERCENTAGE	REFERENCE GUIDE
SWITCH, THERMAL	PARAMETER CHANGE CONTACT FAILURE (OPEN) THERMAL SWITCH FAILURE (STUCK) CONTACT FAILURE (SHORT)	63 27/N 8 2/N	FMD-91
TRANSFORMER	SHORTED WINDING OPEN WINDING	80/N 20/N	MIL-HDK-338
TRANSISTOR	SHORT CIRCUIT OPEN CIRCUIT	73 27	FMD-91

3.7. SRA (SRU) BIT ANALYSIS WORKSHEET FORMAT

The BIT Analysis worksheet for each circuit card/module contained in the system includes the following columns of data described below:

The top of the first page of each worksheet includes the following summary information:

(1) WRA (LRU) name

(2) SRA (SRU) name

(3) SRA (SRU) Module part number

(4) SRA (SRU) Module schematic drawing number and revision used to perform the analysis

(5) SRA (SRU) Module reference designator

(6) Total failure rate of the SRA (SRU) module

(7) Total analyzed SRA (SRU) module failure rate (total less excluded)

(8) Failure class legend (refer to Table 1, items 15

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COL	AME	DESCRIPTION
1	Component Ref/Des	List the component reference designator and include any additional Applicable identifiers (subchip designator - U1A - for multiple gate chips, basic function such as Data0-7 or groups of pins) needed to uniquely relate the referenced component to a specific component on the schematic
2	Part Identification	List the component's military or commercial part number.
3	Component Failure Rate	Identify the failure rate of the component from the reliability prediction. If subsections, multiple subsections or multiple output pins of a component are listed separately, then the total reliability failure rate of the component is distributed equally among the subsections and/or pins analyzed including unused functional pins.
4	Failure Mode	Define the specific failure mode of the component under consideration (e.g. open, short, drift, stuck high, stuck "low", etc.). Applicable failure modes and failure rate distributions for each device are to be derived per Table 2, Failure Mode Allocation/ Distribution of Components. Failure mode allocations for ASICS and other large devices are to be coordinated with Customers. Any deviations or additions to Table 2 are subject to Customer approval.
5	Failure Rate per Mode	Define the portion of the component failure rate attributable to the current failure mode based on the number of failure modes assigned to the device and the distributions from Table 2.
6	Failure Classification	Identify the component failure classification per Table 1.
6a	Failure Applicability	Indicate whether or not the failure is excluded from BIT fault detection requirements as defined in the Performance/Procurement Specification.

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7	Circuit Function	Indicates the component function within the functional path.
8	Effects of Failure	Describe the effect the failure mode on the WRA/SRA (LRU/SRU), including no effect of applicable. The effect should be oriented toward an observable effect that would be evident to the pilot/maintainer and thus usable for manual O-level detection/isolation of a failure.
9	Next Level Effects (box)	Describe the effect that the failure mode has on the box in which the module is installed, including no effect if applicable.
10	Pilot Effects/Observations (system) (O-level Detection)	Identify any effect that the failure would have on system performance observed by the end user. Note that this is especially critical for failures which cannot be detected/isolated by BIT alone.
11	Power-up/Start-up BIT Test Identifier(s)	Identify the applicable power-up/start-up BIT test number which detects the failure, “None” if the fault is not detected in this mode, or “N/A” if the fault is classified as “no effect”. Where more than one test applies, each test is delimited by a semi-colon (;).
12	Background/Periodic BIT Test Identifier(s)	Identify the applicable background/periodic BIT test number which detects the failure, “None” if the fault is not detected in this mode, or “N/A” if the fault is classified as “no effect”. Where more than one test applies, each test is delimited by a semi-colon (;).
13	Initiated BIT Test Identifier(s)	Identify the applicable initiated BIT test number which detects the failure, “None” if the fault is not detected in this mode, or “N/A” if the fault is classified as “no effect”. Where more than one test applies, each test is delimited by a semi-colon (;).

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14	Maintenance BIT Test Identifier(s)	Identify the applicable maintenance BIT test number which detects the failure, “None” if the fault is not detected in this mode, or “N/A” if the fault is classified as “no effect”. Where more than one test applies, each test is delimited by a semi-colon (;).
15	Detected Failure Rate via PUBIT	Indicate whether or not the failure is detectable by PUBIT by listing the associated failure rate from column 5. (Additional columns may be added for other BIT modes, if required.)
16	Detected Failure Rate via PBIT	Indicate whether or not the failure is detectable by PBIT by listing the associated failure rate from column 5.
17	Detected Failure Rate via IBIT	Indicate whether or not the failure is detectable by IBIT by listing the associated failure rate from column 5.
18	Detected Failure Rate via MBIT	Indicate whether or not the failure is detectable by MBIT by listing the associated failure rate from column 5.
19	Detected Failure Rate via IBIT + PBIT	Indicate whether or not the failure is detectable by IBIT + PBIT by listing the associated failure rate from column 5.
20	Detected Failure Rate by the Combined BIT Modes	Indicate whether or not the failure is detectable by any of the combined BIT modes by listing the associated failure rate from column 5.
21	Powerup/ Startup BIT Fault Isolation	Indicate any other aircraft replaceable items that cannot be isolated from the current failure mode by BIT, or “none” as applicable.
22	Background/ Periodic BIT Fault Isolation	Indicate any other aircraft replaceable items that cannot be isolated from the current failure mode by BIT, or “none” as applicable.

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23	Initiated BIT Fault Isolation	Indicate any other aircraft replaceable items that cannot be isolated from the current failure mode by BIT, or “none” as applicable.
24	Maintenance BIT Fault Isolation	Indicate any other aircraft replaceable items that cannot be isolated from the current failure mode by BIT, or “none” as applicable.
25	1 WRA (LRU) Isolation via Combined BIT Modes	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via BIT by listing the associated failure rate from column 5. (Additional columns may be added for the isolation of 2 Weapon Replaceable Assemblies (WRAs) / Line Replaceable Units (LRUs) via BIT, if specified in program requirements.)
25a	1 WRA (LRU) Isolation via PUBIT Modes	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via PUBIT by listing the associated failure rate from column 5.
25b	1 WRA (LRU) Isolation via MBIT Modes	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via MBIT by listing the associated failure rate from column 5.
25c	1 WRA (LRU) Isolation via IBIT	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via IBIT by listing the associated failure rate from column 5.
25d	1 WRA (LRU) Isolation via PBIT	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via PBIT by listing the associated failure rate from column 5
25e	1 WRA (LRU) Isolation via PBIT + IBIT	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via BIT by listing the associated failure rate from column 5
26	Detected BIT Failure Rate Isolated to 1 SRA (SRU)	Indicates whether or not BIT can fault isolate failure to one (1) SRA (SRU) listing the associated failure rate of the component from column 5

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26a	Detected BIT Failure Rate Isolated to 2 SRA (SRU)s	Indicates whether or not BIT can fault isolate failure to two (2) SRA (SRU) listing the associated failure rate of the component from column 5
26b	Detected BIT Failure Rate Isolated to 3 SRA (SRU)s	Indicates whether or not BIT can fault isolate failure to three (3) SRA (SRU) listing the associated failure rate of the component from column 5
27	Analyzed Failure Rate (BIT)	List failure rate from column 5, if normal system operational performance is impacted by failure mode (Failure class 1, 2, or 3) and the failure mode is not excluded from BIT detection (column 6a)
27a	Analyzed Failure Rate (Verticality)	List failure rate from column 5, if normal system operational performance (Failure class 1, 2, or 3) is impacted by failure mode.
27b	BIT Indication	List any BIT error fault Code(s) generated as a result of the failure mode.
28	BIT test information	For analog type test/measurements list the BIT test(s) expected value and tolerance, such as 5v +.2 vdc. For digital type test/measurements list specific pertinent parameters such as “Block 1 memory read/write using AAAA and 5555 patterns” or “verify output went to commanded state” Include methodology for determining tolerance versus equipment performance requirements.

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29	WRA (LRU) Depot test information	List the WRA (LRU) depot level test expected value and tolerance, as applicable. List information similar to column 15 or 17 that would enable a comparison of test/measurement parameters and tolerance. Include methodology for determining tolerance versus next higher maintenance level.
30	Shop Level Test Identifier(s)	List the corresponding WRA (LRU) that will detect the failure in the target repair shop.
31	ATP Test Identifier	List the corresponding WRA (LRU) that will detect the failure in the target repair shop.

At the bottom of columns 5, 15 – 20, and 28 – 30 indicate the total for each column. This total indicates the ability of BIT to detect failures within the Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) and isolate to the correct WRA (LRU). At the bottom of columns 8, 10, 30a, and 32 indicate the total for each column; this total will be used to assess the verticality of the testing methods.

3.8. WRA (LRU) BIT ANALYSIS WORKSHEET FORMAT

Each WRA (LRU) BIT/Verticality worksheet is a summary of the SRA (SRU) BIT Analysis Worksheets. The Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) BIT/Verticality Analysis worksheet includes the following columns of data described below for each Shop Replaceable Assembly (SRA) / Shop Replaceable Unit (SRU):

At the top of the first page of each WRA / LRU worksheet, include the following summary information:

a) WRA (LRU) name
b) WRA (LRU) Assembly number (part number)
c) WRA (LRU) Reference Designator

COL	NAME	DESCRIPTION
1	SRA (SRU) Ref/Des	List the SRA (SRU) reference designator for each SRA (SRU).
2	SRA (SRU) Name	List the SRA (SRU) nomenclature for each SRA (SRU).
3	SRA (SRU) Part Number	List the SRA (SRU) part number for each SRA (SRU).
4	Total SRA (SRU) Failure Rate	List the total SRA (SRU) failure rate by summing column 5 of the appropriate SRA worksheet..

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5	Total Analyzed (BIT) SRA (SRU) Failure Rate	List the total SRA (SRU) failure rate by summing column 5 entries for all non-excluded faults (identified in column 6) of the appropriate SRA (LRU) worksheet.
6	Detected Failure Rate by PowerUp/ Startup BIT detection (PUBIT)	Indicate the total failure rate for the module that is detectable by powerup/ startup BIT, by summing column 5 entries for all nonexcluded faults which are detected by a powerup/ startup BIT test (identified in columns 6 and 11) in the appropriate module worksheet.
7	Detected Failure Rate by Background/Periodic BIT (PBIT)	Indicate the total failure rate for the module that is detectable by background/periodic BIT, by summing column 5 entries for all nonexcluded faults which are detected by a background/ periodic BIT test (identified in columns 6 and 12) in the appropriate module worksheet.
8	Detected Failure Rate by Initiated BIT (IBIT)	Indicate the total failure rate for the module that is detectable by initiated BIT, by summing column 5 entries for all nonexcluded faults which are detected by an initiated BIT test (identified in columns 6 and 13) in the appropriate module worksheet.
9	Detected Failure Rate by Maintenance BIT (MBIT)	Indicate the total failure rate for the module that is detectable by maintenance BIT, by summing column 5 entries for all nonexcluded faults which are detected by a maintenance BIT test (identified in columns 6 and 14) in the appropriate module worksheet.
10	Total BIT Detection	Indicate the total rate for the SRA (SRU) that is detectable by the sum of all BIT modes, by summing column 5 entries for all nonexcluded faults which are detected by any BIT test (identified in columns 6, 15, 16, 17, 18, 20) in the appropriate SRA worksheet.

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11	1 WRA (LRU) Isolated via Combined BIT Mode	Indicate the total SRA (SRU) failure rate that is isolated by BIT to the correct WRA (LRU). (Additional column should be added for 2 WRAs (LRUs) Isolation via BIT, if specified in program requirements.)
11a	1 WRA (LRU) Isolation via PUBIT Modes	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via PUBIT by listing the associated failure rate from column 5.
11b	1 WRA (LRU) Isolation via MBIT Modes	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via MBIT by listing the associated failure rate from column 5.
11c	1 WRA (LRU) Isolation via IBIT	Indicate whether or not the failure is correctly isolated to one (1) WRA (LRU) via IBIT by listing the associated failure rate from column 5.
11d	1 WRA (LRU) Isolation via PBIT	Indicate the total SRA (SRU) failure rate that is isolated by PBIT to the correct WRA (LRU).
11e	1 WRA (LRU) Isolation via PBIT + IBIT	Indicate the total SRA (SRU) failure rate that is isolated by PBIT + IBIT to the correct WRA (LRU).
12	Detected BIT Failure Rate Isolated to 1 SRA (SRU)	Indicate the total module failure rate that is isolated by BIT to the correct module by summing column 5 entries for all nonexcluded faults which are isolated to one module (identified in columns 27 – 27b) in the appropriate module worksheet.
12a	Detected BIT Failure Rate Isolated to 2 SRAs (SRUs)	Indicate the total module failure rate that is isolated by BIT to a maximum of 2 modules by summing column 5 entries for all nonexcluded faults which are isolated to one module (identified in columns 27 – 27b) in the appropriate module worksheet.

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12b	Detected BIT Failure Rate Isolated to 3 SRAs (SRUs)	Indicate the total module failure rate that is isolated by BIT to a maximum of 3 modules by summing column 5 entries for all nonexcluded faults which are isolated to one module (identified in columns 27 – 27b) in the appropriate module worksheet.
13	Detected PBIT + IBIT Failure Rate Isolated to 1 SRA (SRU)	Indicates the total SRA (SRU) failure rate that is isolated by PBIT + IBIT to one (1) SRA (SRU).
13a	Detected PBIT + IBIT Failure Rate Isolated to 2 SRAs (SRUs)	Indicates the total SRA (SRU) failure rate that is isolated by PBIT + IBIT to two (2) SRAs (SRUs).
13b	Detected PBIT + IBIT Failure Rate Isolated to 3 SRAs (SRUs)	Indicates the total SRA (SRU) failure rate that is isolated by PBIT + IBIT to three (3) SRAs (SRUs).
14	Comments	If the failure is isolated to more than one (1) WRA (LRU), list external WRAs (LRUs) within that group.
15	Total Analyzed (Verticality) SRA (SRU) Failure Rate	List the total analyzed (electronic component) failure rate for each SRA (SRU)
16	Total Overall BIT/O level fault detection	List the failure rate detected by BIT/O-level troubleshooting procedures for each SRA (SRU).
17	Shop-Level Fault Detection	Indicate the total failure rate for the module that is detectable in the target repair shop, by summing column 5 entries for all nonexcluded faults which are detected by depot level tests (identified in columns 6 and 21) in the appropriate module worksheet.
18	ATP Verticality	Indicate the total failure rate for the module that is detectable by the ATP, by summing column 5 entries for all nonexcluded faults which are detected by ATP tests (identified in columns 6 and 22) in the appropriate module worksheet.

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At the bottom of columns 4 through 18 indicate the total for each column. These totals indicate the ability of Built-In Test (BIT) to detect failures within the Weapon Replaceable Assembly (WRA) / Line Replaceable Unit (LRU) and isolate to the correct WRA / LRU. At the bottom of columns 13 through 16, indicate the total for each column. These totals are used to indicate the overall verticality of the testing methods.

3.9. SYSTEM BIT/VERTICALITY ANALYSIS SUMMARY SHEET FORMAT

The System Level BIT/Verticality Analysis Summary Sheet is a summary of the performance of Weapon Replaceable Assemblies (WRAs) / Line Replaceable Units (LRUs). The System BIT/Verticality Analysis Summary Sheet contains the following columns of data as described below. For systems with multiple WRAs / LRUs, provide a system level BIT/Verticality analysis summary sheet in a similar manner:

COL	NAME	DESCRIPTION
1	% Analyzed Failure Rate Detected by Power-Up Built-In Test (PUBIT)	Calculate the percentage of analyzed failure rate detected by PUBIT. (Additional columns should be added for other BIT modes, if required.)
2	% Analyzed Failure Rate Detected by Maintenance-Up Built-In Test (MUBIT)	Calculate the percentage of analyzed failure rate detected by MUBIT. (Additional columns should be added for other BIT modes, if required.)
3	% Analyzed Failure Rate Detected by Periodic Built-In Test (PBIT)	Calculate the percentage of analyzed failure rate detected by PBIT.
4	% Analyzed Failure Rate Detected by Initiated Built-In Test (IBIT)	Calculate the percentage of analyzed failure rate detected by IBIT.
5	% Analyzed Failure Rate Detected by PBIT + IBIT	Calculate the percentage of analyzed failure rate detected by PBIT + IBIT.
6	% Analyzed Failure Rate Detected by the Combined BIT Modes	Calculate the percentage of analyzed failure rate detected by the combined BIT modes.
7	% Detected Failure Rate Isolated to 1 WRA (LRU) by BIT	Calculate the percentage of detected failure rate unambiguously isolated to the correct WRA (LRU) by BIT. (Additional column should be added for 2 WRAs (LRUs) Isolation via BIT, if specified in program requirement

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8a	% Detected Failure Rate Isolated to 1 SRA (SRU) by PBIT + IBIT	Calculated the percentage of detected failure rate isolated to the correct SRA (SRU)
8b	% Detected Failure Rate Isolated to </= 2 SRA (SRU) by PBIT + IBIT	Calculated the percentage of detected failure rate isolated to two (2) or less SRAs (SRUs)
8c	% Detected Failure Rate Isolated to </= 3 SRA (SRU) by PBIT + IBIT	Calculated the percentage of detected failure rate isolated to three (3) or less SRAs (SRUs)
9	% Detected Failure Rate at the O-Level (Verticality)	Calculate the percentage of analyzed failure rate (Verticality) detected by BIT/O-level methods.
10	% Detected Failure Rate at the depot level (Verticality)	Calculate the percentage of analyzed failure rate (Verticality) detected at the WRA (LRU) depot level of testing.

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11	% Detected Failure Rate using the ATP	Calculate the percentage of analyzed failure rate (rate used for verticality) that the ATP could detect.

3.10. Failure Mode and Effects Analysis FMEA. For programs where a separate Failure Mode and Effects Analysis (FMEA) or Failure Mode, Effects, and Criticality Analysis (FMECA) System Design Language (SDL) item is not ordered, this section fulfills the intent of the Failure Mode and Effects Analysis (E-14.04/E-49), required for support of System Safety Engineering (System Safety Hazard Analysis) and Product Support Engineering (Avionics Maintenance Information Data System (AMIDS) or Configuration and Fault Reporting System (CF/CRS) and Tech Pubs development). This analysis extends the Built-In Test/Test Verticality (BIT/TV) Analysis and provides for the identification and classification of failure modes and safety hazards per the detailed ground rules provided herein. For each component failure mode of the avionics system analyzed in section A, the FMEA analysis includes the following information. In addition to the component failure modes identified in the BIT Analysis, the FMEA also addresses external failure modes and failure effects of damaged Weapon Replaceable Assembly (WRA) connectors, aircraft wiring, and associated airborne equipment. For this analysis, fault detection and isolation coverage by all available means to the operator/maintainer (including BIT) is identified. Provided that the BIT/TV analysis is used as its baseline, this section may be tailored to a functional analysis, subject to Customer written approval.

3.10.1. WRA/LRU Reference Designator (per BIT Analysis).

3.10.2. SRA/SRU Reference Designator (per BIT Analysis).

3.10.3. Component Reference Designator (per BIT Analysis).

3.10.4. Circuit Function (per BIT Analysis).

3.10.5. Failure Mode (per BIT Analysis).

3.10.6. Mission Phase/Operational Mode (per MILSTD1629A, Task 101).

3.10.7. SRA/SRU (local) Failure Effect (per MILSTD1629A, Task 101).

3.10.8. WRA/LRU (next level) Failure Effect (per MILSTD1629A, Task 101).

3.10.9. Aircraft (end) Failure Effect (per MILSTD1629A, Task 101).

3.10.10. Failure Detection/Isolation Method (see BIT Analysis & MILSTD1629A, Tasks 101 &103).

3.10.11. Compensating Provisions (per MILSTD1629A, Task 101).

3.10.12. Severity Classification (per MILSTD1629A, Task 101).

3.10.13. Remarks (per MILSTD1629A, Task 101).

3.10.14. Failure Probability/Failure Rate Data Source (per MILSTD1629A, Task 102).

3.10.15. Failure Effect Probability (per MILSTD1629A, Task 102).

3.10.16. Failure Mode Ration (per MILSTD1629A, Task 102).

3.10.17. Operating Time (per MILSTD1629A, Task 102).

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3.10.18. Engineering Failure Mode MTBF (per MILSTD1629A, Task 103).

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